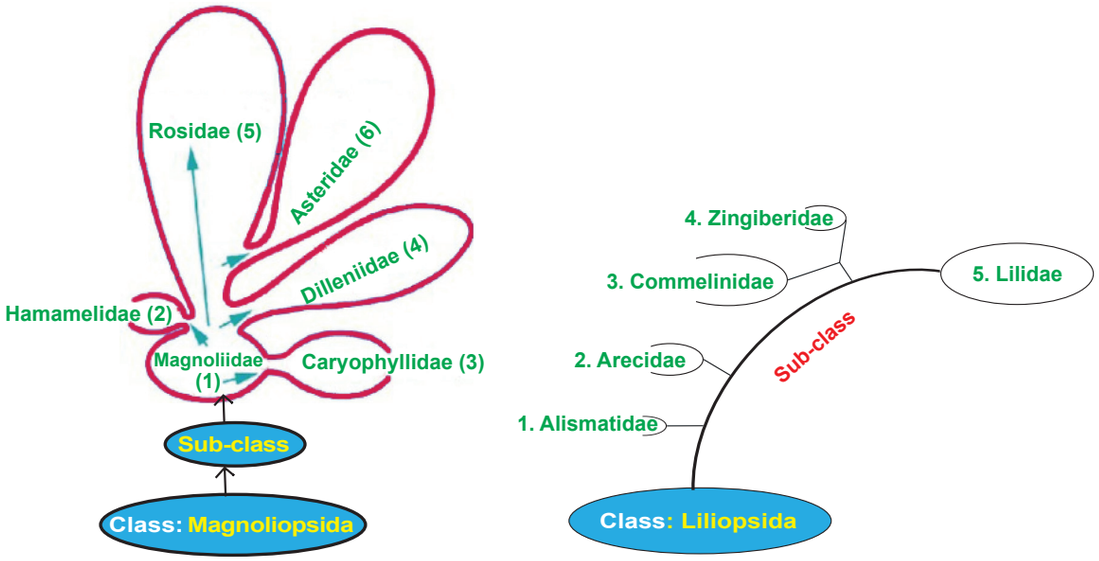


Figure 5.10: Diagramatic representation of class Magnoliopsida and Liliopsida.

genes (*atp*B and *rbc*L) and one nuclear gene (nuclear ribosomal 18s DNA).

The most recent updated version, APG IV (2016) recognised 64 orders and 416 families. Of these, 416 families 259 are represented in India.

The outline of APG IV classification is given below.

Angiosperms are classified into three clades early angiosperms, monocots and eudicots. Early angiosperms are classified into **8 orders** and **26 families** (**ANA**-clade + magnoliids + Chloranthales)

Amorbellales
Nymphaeales
Austrobaileyales

- Seeds generally always with two cotyledons.
- Presence of ethereal oils.
- Leaves are always simple net-veined.
- Each floral whorls with many parts.
- Perianth usually spirally arranged or parts in threes.
- Stamens with broad filaments.
- Anthers tetrasporangiate.
- Pollen monosulcate.
- Nectaries are rare.
- Carpels usually free and
- Embryo very small.

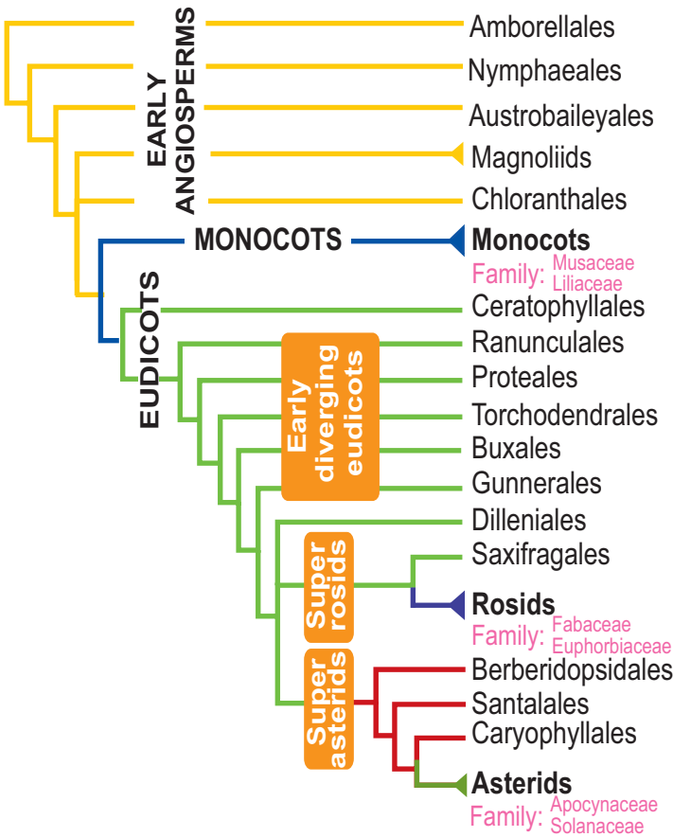


Figure 5.11: Simplified version of APG IV
(Source: Plant Gateway's The Global Flora, Vol. I January 2018)

Monocots are classified into 11 orders and 77 families (basal monocots + lilioids + commelinids)

- Seeds with single cotyledon.
- Primary root short-lived.

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